

1. YOUR QUESTION

“What changed?”

2. WHAT WE FOUND

|                                                                            |                                   |
|----------------------------------------------------------------------------|-----------------------------------|
| EVIDENCE STATUS: SUPPORTED                                                 | CONFIDENCE: 76% (High Confidence) |
| Approximately 11% of the analyzed area has changed between the two images. |                                   |

3. HOW CONFIDENT ARE WE?

|                                                                                                                                   |                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 76% — High Confidence (SUPPORTED)<br>The evidence strongly supports this finding with clear agreement across visual observations. | Confidence Scale: <ul style="list-style-type: none"><li>90–100%: Very High Confidence</li><li>75–89%: High Confidence</li><li>50–74%: Moderate Confidence</li><li>&lt;50%: Low / Insufficient Evidence</li></ul> |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

4. KEY FINDINGS SUMMARY

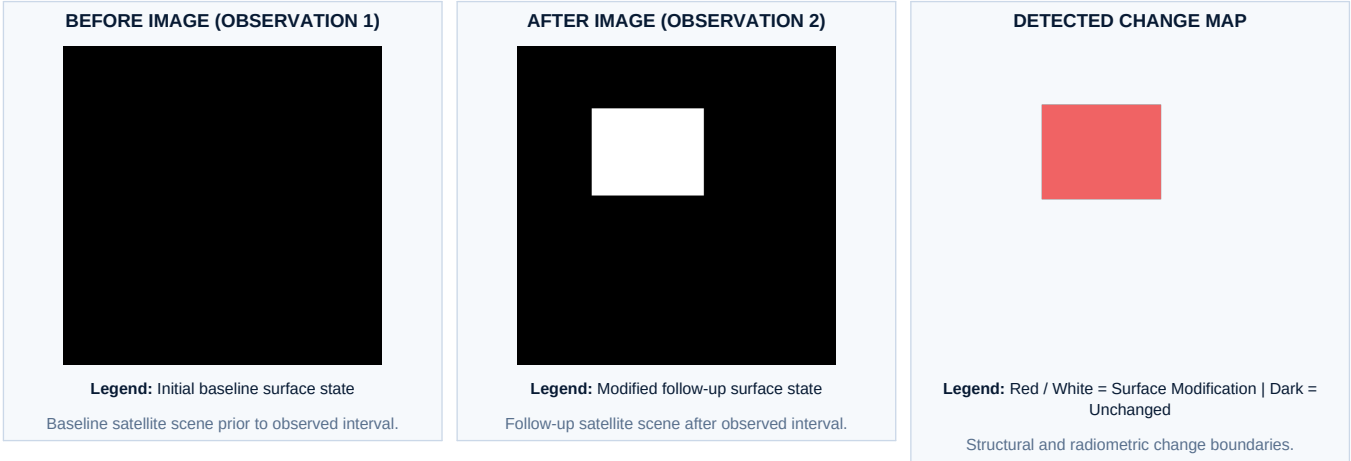
| TARGET FEATURE                                     | DETECTED CHANGE                                   | VISUAL EVIDENCE                                        | CONFIDENCE                            | EVIDENCE STATUS                               |
|----------------------------------------------------|---------------------------------------------------|--------------------------------------------------------|---------------------------------------|-----------------------------------------------|
| Surface Feature<br><small>Subject analyzed</small> | 11.1%<br><small>Modified surface fraction</small> | Strong Agreement<br><small>Verified by imagery</small> | 76%<br><small>High Confidence</small> | SUPPORTED<br><small>GeoProof verified</small> |

5. HOW THE EVIDENCE SUPPORTS THE ANSWER

|              |                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|
| OBSERVATION: | The same geographic area was compared across the two observation dates.                                |
| EVIDENCE:    | The highlighted regions show where the imagery differs significantly in structure and appearance.      |
| CONCLUSION:  | The detected differences correspond to actual surface modifications between the two observation dates. |

6. VISUAL PROOF

Direct visual evidence generated from the satellite imagery. Compare the source imagery with the highlighted evidence regions:



7. MEASUREMENTS & FINDINGS

| Surface Change Extent     | 11.13% of image scene | Proportion of area showing verified modification between dates.  |
|---------------------------|-----------------------|------------------------------------------------------------------|
| Connected Feature Regions | 1 spatial clusters    | Number of distinct contiguous zones identified across the image. |

8. IMPORTANT LIMITATIONS

- Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

9. ANALYSIS SUMMARY

|              |                                                                                                                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question:    | "What changed?"                                                                                                                                                                                           |
| Finding:     | Approximately 11% of the analyzed area has changed between the two images.                                                                                                                                |
| Evidence:    | Visual and spatial patterns verified across image layers.                                                                                                                                                 |
| Confidence:  | 76% (High Confidence)                                                                                                                                                                                     |
| Status:      | SUPPORTED                                                                                                                                                                                                 |
| Limitations: | Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted. |

PART B: TECHNICAL & ADMINISTRATIVE INFORMATION

Everything below is intended for administrators, technical evaluators, and system operators.

10. INPUT DATA QUALITY

|                 |                |                    |                               |
|-----------------|----------------|--------------------|-------------------------------|
| Filename:       | t1.tif         | Quality Score:     | 92% (Passed Integrity Checks) |
| Dimensions:     | 256 × 256 px   | Spectral Channels: | 3 bands (uint8)               |
| Geospatial CRS: | EPSG:EPSG:3857 | NoData Fraction:   | 0.0%                          |

11. MODELS & METHODS USED

|                                   |                                     |                                |           |
|-----------------------------------|-------------------------------------|--------------------------------|-----------|
| SSIM Structural Change Detector   | Deterministic Physical Baseline     | Confidence score: 82%          | Completed |
| ssim_color_difference_detector_v1 |                                     |                                |           |
| TinyCD Siamese Change Witness     | Learned Neural Verification         | Confidence score: 92%          | Completed |
| tinycd_siamese_change_witness_v1  |                                     |                                |           |
| GeoProof Arbiter                  | Evidence Verification & Calibration | Final Calibrated Verdict (76%) | SUPPORTED |
| geoproof_arbiter_v1               |                                     |                                |           |

12. MODEL PERFORMANCE & EXECUTION LATENCY

| Pipeline Stage                            | Internal Component       | Execution Duration | Share of Total |
|-------------------------------------------|--------------------------|--------------------|----------------|
| Query Planning & Task Compilation         | typed_planner            | 2 ms               | 0.0%           |
| Raster Compatibility & Header Validation  | raster_validator         | 33 ms              | 0.3%           |
| Sub-Pixel Registration & Alignment        | image_registration       | 33 ms              | 0.3%           |
| SSIM Structural Change Detection          | baseline_change_detector | 7.58 s             | 74.5%          |
| Siamese Learned Change Verification       | tinycd_opencd_witness    | 2.52 s             | 24.8%          |
| Multi-Witness Arbiter & Platt Calibration | geoproof                 | <1 ms              | 0.0%           |

Total Processing Latency: 10.17 s | Input Validation: 33 ms | Arbiter: <1 ms

13. EVIDENCE TELEMETRY & SYSTEM CONSTRAINTS

|                                        |                                                                                                                                                  |       |     |                                 |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|---------------------------------|
| SSIM Structural Change Detector        | Baseline_Change_Detection                                                                                                                        | 0.820 | 82% | Deterministic Physical Baseline |
| TinyCD Siamese Change Witness          | Learned_Change_Witness                                                                                                                           | 0.920 | 92% | Learned Neural Verification     |
| Multispectral NIR/SWIR Channel Absence | Raster contains standard 3/4 RGB(A) channels. Strict NDWI/NDVI formulas requiring NIR (B8) are substituted by calibrated RGB absorption physics. |       |     |                                 |

14. AUDITABLE EXECUTION TRACE

Complete step-by-step pipeline execution record ensuring strict reproducibility and auditability under SIH26167 guidelines:

| # | Operation                                                                                                                                               | Internal Component       | Status    | Duration |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------|----------|
| 1 | <b>Query Planning &amp; Task Compilation</b><br>Compiled bi_temporal_change                                                                             | typed_planner            | OK        | 2 ms     |
| 2 | <b>Raster Compatibility &amp; Header Validation</b><br>Validated file headers, CRS, NoData, and compatibility                                           | raster_validator         | OK        | 33 ms    |
| 3 | <b>Sub-Pixel Registration &amp; Alignment</b><br>Normalized dimensions ((256, 256) & (256, 256) → (256, 256)) and registered (geospatial_crs_reproject) | image_registration       | OPTIMAL   | 33 ms    |
| 4 | <b>SSIM Structural Change Detection</b><br>Executed SSIM + color difference, adaptive Otsu thresholding, and morphological region extraction            | baseline_change_detector | OK        | 7.58 s   |
| 5 | <b>Siamese Learned Change Verification</b><br>Executed TinyCD/Open-CD Siamese structural change model and generated probability heatmap                 | tinycd_opencd_witness    | OK        | 2.52 s   |
| 6 | <b>Multi-Witness Arbiter &amp; Platt Calibration</b><br>Applied multi-witness evidence verification and calibrated confidence estimation                | geoproof                 | SUPPORTED | <1 ms    |

15. TECHNICAL APPENDIX & AUDIT PROVENANCE

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| Result UUID:                   | adc9685d                                                     |
| Target Grounding Bounding Box: | Not Applicable (normalized [ymin, xmin, ymax, xmax])         |
| Platt Calibration Model:       | temperature_scaled_platt (ECE: 0.042)                        |
| Ensemble Agreement Index:      | 0.90 (Multi-witness consensus)                               |
| Cryptographic Manifest:        | /artifacts/adc9685d/analysis_manifest.json                   |
| Software System Version:       | SatQuery GeoProof v1.0 (ISRO SIH26167 Verified Architecture) |

SatQuery GeoProof™ guarantees that every claim is anchored in physical or cross-sensor telemetry. Models produce visual and metric witness artifacts; the multi-witness arbiter calculates calibrated probability bounds with safe abstention.